GENE CARLO FRANCO BELINARIO

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Interests

- **Astrophysics**, **Cosmology**, **Computational Physics**, **Physics Education**
- Galaxy Clusters, Gravitational Waves, Gravitational Lensing, Dark Energy, Data Analysis, Machine Learning

Education

- **Master of Science in Physics**
  Feb 2023 – July 2025
  GWA: 1.5700 (1.00 highest)
  National Institute of Physics
  University of the Philippines
- **Bachelor of Science in Applied Physics**
  Aug 2018 – Feb 2023
  GWA: 1.2843 (1.00 highest), *Magna Cum Laude*
  National Institute of Physics
  University of the Philippines

Research Experience

- **Impact of Incomplete Quenching on Red-Sequence Cluster Selection beyond $z = 1$: Conditional Richness Scatter and Selection-Induced Assembly Bias**
  Remote Collaboration • Aug 2026 – Present
  Collaborator: T. Sunayama
  - Compared halos selected by richness of red-member galaxies and by richness of all galaxies in the UCHUU-UNIVERSEMACHINE galaxy catalog at $z = 1.00 - 2.00$, measuring the red fraction, the scatter at fixed halo mass, and the selection completeness.
  - Measured halo auto- and cross-correlation functions of the selected samples, and tested their bias against DARK EMULATOR predictions and against mass-matched and shuffled control samples to isolate selection-induced halo assembly bias.

- **Finding Blue Clusters: When did Galaxy Clusters Stop Forming Stars?**
  Institute of Astronomy and Astrophysics, Academia Sinica • June – Aug 2026
  Supervisor: T. Sunayama
  - Identified galaxy cluster candidates in the UCHUU-UNIVERSEMACHINE galaxy catalogs across multiple snapshots.
  - Investigated the evolution of star-forming and quenched clusters and their dependence on redshift, cluster mass, and environment.
  - Found that the red galaxies and clusters dominate at $z \sim 0.50$, driven mainly by redshift rather than cluster mass.

- **Cosmo-Learn: Code for Learning Cosmology using Different Methods and Mock Data**
  National Institute of Physics, University of the Philippines Diliman • June 2025 – July 2026
  Collaborators: R.C. Bernardo, D. Grandón, J.L. Said, V.H. Cárdenas, and R. Reyes
  - Co-developed `cosmo-learn`, a Python toolkit that simulates cosmological data and benchmarks statistical and machine learning inference methods.
  - Implemented the simulation pipeline that generates bright standard sirens under various cosmological models and supermassive black hole binary populations.

- **Constraining Cosmic Gravity with Gravitational-Wave Standard Sirens**
  M.Sc. Thesis • National Institute of Physics, University of the Philippines Diliman • Feb 2023 – July 2025
  Supervisors: R.C. Bernardo, R. Reyes, and I. Vega
  - Simulated mock bright standard siren catalogs under various cosmological models using LISA forecasts on supermassive black hole binary populations.
  - Inferred cosmological parameters using standard sirens independently and jointly with Type Ia supernovae (SNeIa), baryon acoustic oscillations (BAO), and cosmic chronometers (CC).
  - Quantified how far sirens alone are biased and weakly constraining, and showed they marginally tighten joint estimates with SNeIa + BAO + CC.

Research Experience (continued)

- **Gravitational Lensing Signatures in Gravitational-Wave Signals from Compact Binary Coalescences**
  B.Sc. Thesis • National Institute of Physics, University of the Philippines Diliman • Aug 2020 – Feb 2023
  Supervisor: R. Reyes
  - Showed that galaxy lensing combined with microlensing makes GW signals detectable in the LIGO detectors and inflates the inferred microlens mass by $2-3$ orders of magnitude.

Publications

Journal Articles

1. R. C. Bernardo, D. Grandón, J. L. Said, V. H. Cárdenas, G. C. Belinario, and R. Reyes, "Cosmo-learn: Code for learning cosmology using different methods and mock data," *The European Physical Journal C*, 2026. DOI: https://doi.org/10.1140/epjc/s10052-026-15981-w.
2. E. Di Valentino, J. L. Said, G. C. Belinario, and The CosmoVerse Network, "The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics," *Physics of the Dark Universe*, 2025. DOI: https://doi.org/10.1016/j.dark.2025.101965.

Conference Proceedings

1. G. C. Belinario, R. C. Bernardo, and R. Reyes, "Constraining cosmic dark energy using bright standard sirens," in *Proceedings of the Samahang Pisika ng Pilipinas*, vol. 42, 2024, SPP-2024-2E–01. URL: https://proceedings.spp-online.org/article/view/SPP-2024-2E-01.
2. J. Garrido, G. C. Belinario, J. Rodelas, and J. P. Esguerra, "Lessons from the First Philippine Junior Physics Challenge Examination," in *Proceedings of the Samahang Pisika ng Pilipinas*, vol. 42, 2024, SPP-2024-1E–01. URL: https://proceedings.spp-online.org/article/view/SPP-2024-1E-01.
3. G. C. Belinario and R. Reyes, "Gravitational lensing signatures in gravitational waves from binary black hole mergers," in *Proceedings of the Samahang Pisika ng Pilipinas*, vol. 41, 2023, SPP-2023-1G–04. URL: https://proceedings.spp-online.org/article/view/SPP-2023-1G-04.

In Preparation

1. G. C. Belinario and T. Sunayama, "Impact of incomplete quenching on red-sequence cluster selection beyond $z = 1$: Conditional richness scatter and selection-induced halo assembly bias," in preparation for Physical Review D.

Participations

- **AstroAI Asian Network Summer School 2026**, 24-28 August 2026, *Summer School* by the Academia Sinica Institute of Astronomy and Astrophysics (ASIAA) & Leung Center for Cosmology and Particle Astrophysics (LeCosPA) held at the National Taiwan University, Taiwan. *Project:* "Using Variational Autoencoder for Generating Synthetic Galaxy Spectra"
- **ASIAA Summer Student Program 2026**, 29 June - 21 August 2026, *Research Internship* by the Academia Sinica Institute of Astronomy and Astrophysics (ASIAA) held at the National Taiwan University, Taiwan. *Project:* "Finding Blue Clusters: When did Galaxy Clusters Stop Forming Stars?"
- **APCTP Workshop on Gravitational Waves 2025**, 20-25 April 2025, *Workshop* by the Asia Pacific Center for Theoretical Physics (APCTP) held at the Institute of Physics, Academia Sinica, Taiwan.
- **42nd Samahang Pisika ng Pilipinas Physics Conference and Annual Meeting**, 03-06 July 2024, *Conference* held at the Batangas State University, Philippines. *Contributed Talk:* "Constraining cosmic dark energy with bright standard sirens"
- **Summer School on Cosmology**, 17-28 June 2024, *Summer School* by The Abdus Salam International Center for Theoretical Physics (ICTP) held at Trieste, Italy.
- **LISA Analysis Tools Workshop**, 15-18 April 2024, *Online Workshop* by the LISA Consortium
- **41st Samahang Pisika ng Pilipinas Physics Conference and Annual Meeting**, 19-21 July 2023, *Conference* held at the Del Carmen Convention Center and Bayay Numancia, Philippines. *Contributed Talk:* "Gravitational lensing signatures in gravitational waves from binary black hole mergers"

Participations (continued)

- **6th GW Open Data Workshop**, 15-17 May 2023, *Online Workshop* by the Gravitational Wave Open Science Center (GWOSC)

Skills

- **Programming:** Python, C++, Fortran, Mathematica, Git, Open MPI
- **Computing:** `numpy`, `scipy`, `pandas`, `matplotlib`, `emcee`, `h5py`, `scikit-learn`
- **Astrophysics & Cosmology:** `colossus`, `astropy`, `dark_emulator`, `getdist`, `cosmosis`, `lenstronomy`
- **Gravitational Waves:** `gwpy`, `lalsuite`, `gwosc`, `pycbc`, `bilby`, `lensinggw`
- **Methods:** Correlation measurement, jackknife covariance, parameter estimation, data simulation, machine learning
- **Languages:** English, Filipino

Recognitions and Scholarships

- **Magna Cum Laude**
  Feb 2023
  College of Science
  University of the Philippines
- **Merit Scholar**
  Aug 2018 – July 2022
  Science Education Institute
  Dept. of Science & Technology
- **University Scholar**
  Academic Year 2018 – 2023
  College of Science
  University of the Philippines

Teaching

- **Instructor**
  Sept 2023 – July 2025
  National Institute of Physics
  University of the Philippines
- **Teaching Associate**
  Feb 2023 – Aug 2023
  National Institute of Physics
  University of the Philippines

Involvements

- **Member**, Oct 2024 – Present
  Observational Cosmology and Systematics Working Group, CosmoVerse CA21136 | CosmoVerse COST Action
  Data Analysis in Cosmology Working Group, CosmoVerse CA21136 | CosmoVerse COST Action
- **Head of Technical Services Committee**, Sept 2024 – Jul 2025
  Junior Faculty, National Institute of Physics, University of the Philippines Diliman | NIP
- **Referee**, July 2023 – 2024
  Proceedings of the Samahang Pisika ng Pilipinas | SPP
- **Instructor**, Oct – Dec 2023
  Introduction to Python Programming for Scientific Computing | SEDSPH
- **Member-at-Large** & **Co-founder**, Mar 2021 – Aug 2022
  Students for the Exploration and Development of Space in the Philippines | SEDSPH
- **Vice President for Academic Affairs**, Sept 2020 – July 2021
  UP Physics Association, National Institute of Physics, University of the Philippines Diliman | UPPA

References

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